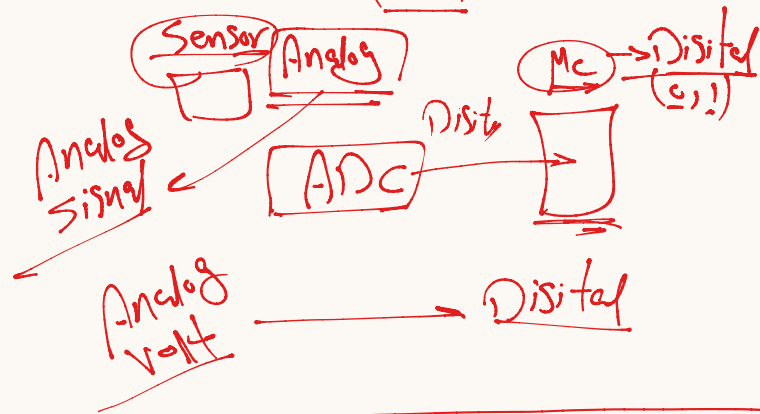
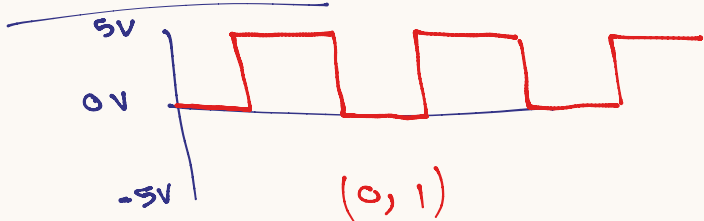


* ADC :: Analog to Digital Convert
 * DAC :: Digital to Analog Convert

* Analog Signal

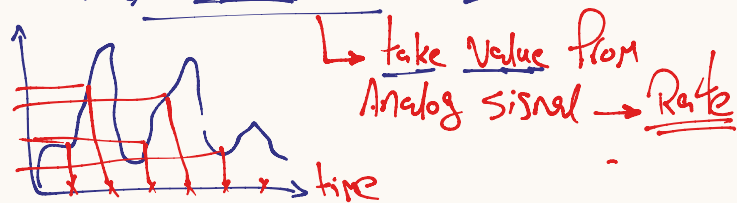


* Digital Signal



→ to Convert from Analog to Digital

→ ① Sampling → Rate



→ ② Quantization ::

→ Convert the value from sampling → to Digital value

→ Resolution
→ Max value (volt)

① Sampling

② Quantization
(ADC)

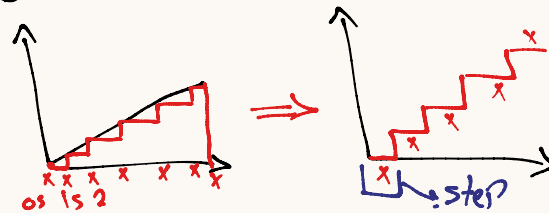
* Quantization → Convert value into digital

→ ① Resolution → # No. of bits
 → Digital value → bits?
 → ② Max volt → Analog →

ex:- System Resolution = 26 bit
 Max volt = 4V



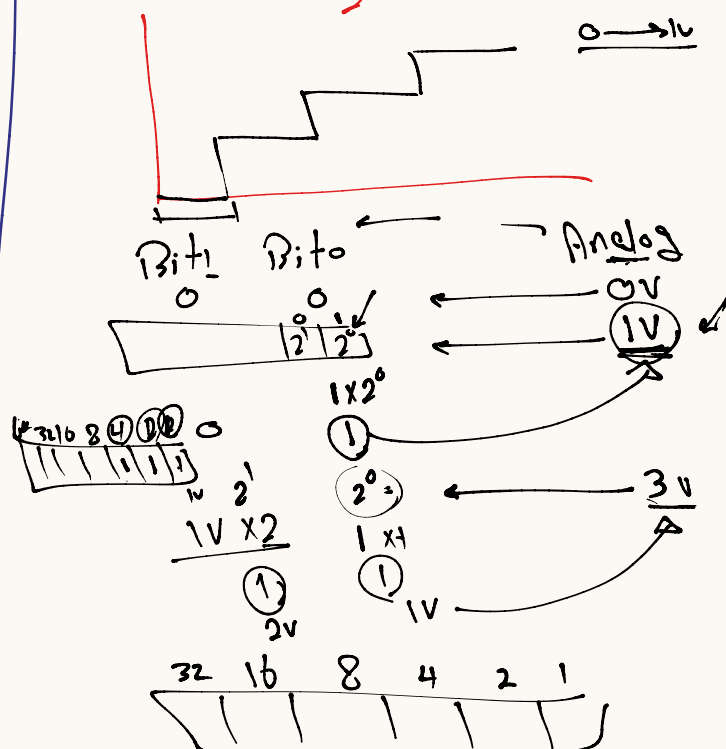
① Sampling & Holding



② Quantization ← Res:- 26 bit
 Max:- 4V

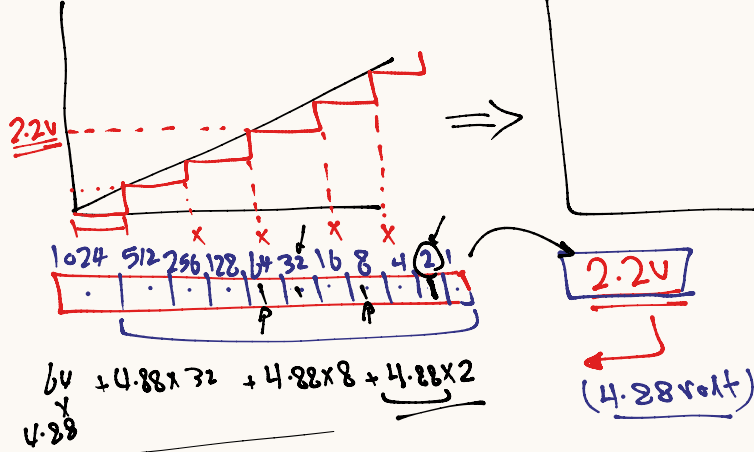
→ step value = $\frac{\text{Max volt}}{\text{Resolution}}$

$$\text{step} = \frac{4}{2^2} = \frac{4}{4} = 1 \text{ volt}$$



ex.: Max Volt = 5V Res 10

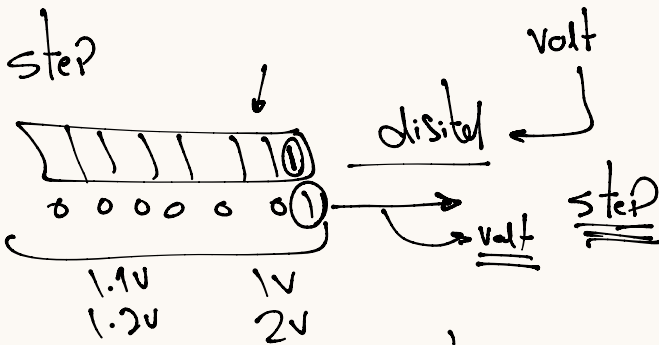
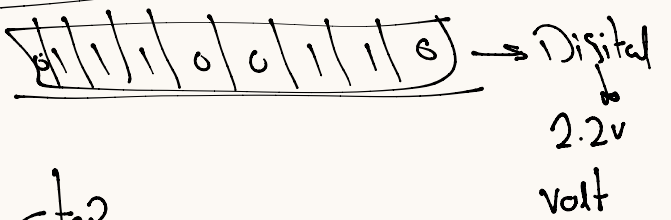
$$\text{Step} = \frac{\text{Max Volt}}{2^{\text{Res}}} = \frac{5}{1024} = \underline{4.88 \text{ mV}} \quad \text{Res } 10$$



$\times \rightarrow$ Quantization error

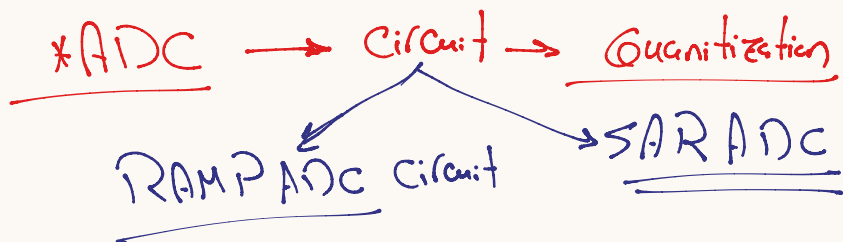
- ① increase Resolution
- ② decrease Max Volt

1.15
1.16
1.17

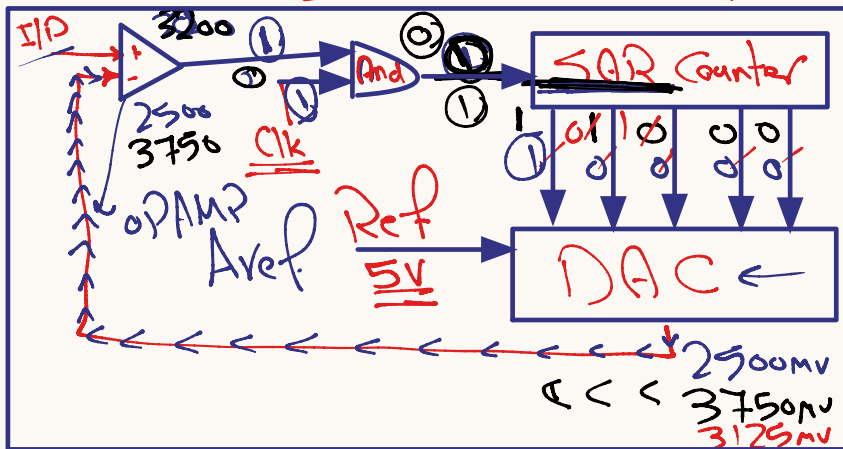


$$\text{Step} \Rightarrow \frac{\text{Max Volt}}{2^{\text{Resolution}}} \rightarrow \uparrow$$

Resolution $\rightarrow \uparrow$



Successive Approximation Register



Digital to Analog Convert

16 8 4 2 1
0 0 0 0 →

$$16 \times 156.25 \text{ mV} = 2500 \text{ mV}$$

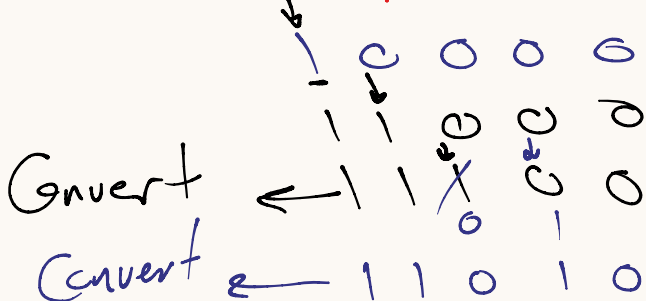
1 1 0 0 0

$$156.25 \text{ mV} \times 16 + 156.25 \text{ mV} \times 8 \Rightarrow 3750$$

1 0 1 0 0

$$156.25 \text{ mV} \times 16 + 156.25 \text{ mV} \times 4 \Rightarrow 3125$$

SAR → MSB → 1



Input 3.2 Res = 5 (10)
Max Ref = 5V

$$\text{Step} = \frac{5 \text{ V}}{2^5} = \frac{5 \times 10^3}{2^5} = 156.25 \text{ mV}$$

① set the MSB b-11

② Convert the Digital to Analog

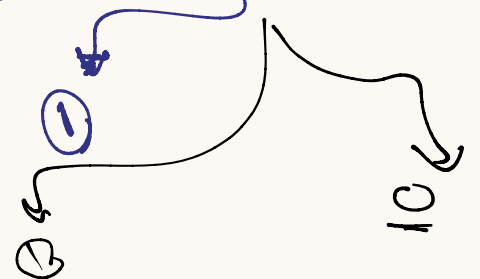
③ Compare with IP b- (COMPARE)

if +ve > -ve
→ ①

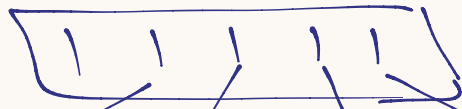
else +ve < -ve
→ 0

④ And. gate

⑤ SAR Counter



Max Convert



$\rightarrow 156.25$

$$+ 156.25 \times 8 + 156.25 \times 4 + 156.25 \times 2 + 156.25 \times 1$$

$$\Leftarrow 156.25 \times 16$$

$$4843.75 \text{ mV}$$

$$5000 - 4843.75 \Rightarrow$$
$$156.25$$

Max Convert $\Rightarrow$ Max Volt-Step